Eda Gjergo, Ph.D.

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Email: eda.gjergo@gmail.com **Publications** **Web**: https://www.edagjergo.com
www.hep.anl.gov/egjergo

Research interests

Galactic chemical evolution, initial mass function, dust modeling, cosmological simulations, interstellar and intracluster medium, supernova cosmology, nonlinear gravity, and relativistic cosmology.

Current employment

Beihang University Beijing, China
Assistant Professor Nov 2026 – Present

Past employment

Nanjing University Nanjing, China
Assistant Research Fellow Sep 2025 – Sep 2026
Nanjing University Nanjing, China
Postdoctoral Scholar in Astronomy Sep 2022 – Aug 2025
Wuhan University Wuhan, China
Postdoctoral Scholar in Physics Oct 2019 – Aug 2022
INAF Trieste Observatory Trieste, Italy
Research Assistant Oct 2015 – May 2019
Argonne National Laboratory Lemont, IL, USA
Junior Team Member Sep 2011 – Aug 2015
Sperling & Kupfer Milan, Italy
Outreach author Jul 2005 – Jul 2007

Education

Università degli Studi di Trieste Trieste, Italy
PhD in Physics Graduated in Spring 2019
Illinois Institute of Technology (IIT) Chicago, IL, USA
BS in Physics Graduated in Spring 2014
Illinois Institute of Technology (IIT) Chicago, IL, USA
BS in Applied Mathematics Graduated in Spring 2014
United World College (UWC) Duino, Italy
International Baccalaureate (IB diploma programme) Spring 2008

Teaching experience

Invited Lecturer (Charles University, Prague, Czech Republic)
Selected topics in Astrophysics, 6 hours Winter 2025-2026
Invited Lecturer (Nanjing University, Nanjing, China)
Introduction to Galactic Chemical Evolution, 2 hours Spring 2023
Invited Lecturer (Wuhan University, Wuhan, China)
Classical Mechanics, 2 hours Fall 2020

Invited Lecturer (University of Trieste, Trieste, Italy)

General Relativity & Cosmology, 2 hours — Spring 2017/2018

Invited Lecturer Yerkes Observatory Summer Program (by the University of Chicago, Chicago, IL, USA)

Introduction to Cosmology, 1 hour — Summer 2013

Honors and scholarships

Natural Science Foundation of China (NSFC) 56k USD — 2026-2027

National Institute of Astrophysics Fellowship (INAF) — 2015-2018

Chambliss Astronomy Achievement Student Award (AAS) — 2013

Merit-based International Scholarship (IIT) — 2008-2014

Winner of the Frascati Scienza National Award — 2010

Italian Ministry of Foreign Affairs Full Scholarship (UWC) — 2006-2008
(3.5% acceptance rate)

Mentoring

5. Soonchul CHOI, Postdoc, IBS, Daejeon, South Korea — 2024-2026

4. Hao-Yang LI, Nanjing University Undergraduate — 2024-2026

3. Zi-Hao LIN, Nanjing University undergraduate –> Nanjing University graduate student. Earned a top-tier grade and a 10,000 RMB research prize for his undergraduate research with me — 2022-2023

2. Run-Dong YU, Nanjing University undergraduate –> Johns Hopkins University graduate student — 2022-2023

1. Jin-Ning LIANG, Wuhan University undergraduate –> Durham University graduate student — 2021-2023

Skills

Public Software I Developed and Released: GalCEM, pyIGIMF

Programming: Python, C, Mathematica, MATLAB, C++, IDL, Fortran90/77.
Specialized software: GADGET-2/3, CosmoMC, SNANA, SNCosmo
Editor/Misc.: Emacs, Vi, SVN, TotalView, Anaconda
Markup: HTML/CSS, LaTeX
Languages: (native) English, Italian, Albanian (beginner) Chinese

Service and outreach

Non-fiction bestseller *Così Parlano Le Stelle* — Apr '07

YouTube Astrophysics Outreach: TheCosmicWeb — Dec '12 – Feb '14

Publications, Codes, and Talks

Monographs, Reviews and Proceedings

3. The relevance of the cosmic microwave background for cosmology

Kroupa P., *Gjergo E.*, Samaras N. 2026

Proceedings of Science, CORFU2025, 224 (2026)

2. The initial mass function of stars

Kroupa P., *Gjergo E.*, Jerabkova T., Yan Z. 2026

Encyclopedia of Astrophysics, Volume 2, ISBN 978-0-443-21439-4, Elsevier, 2026, pp. 173-210

1. The many tensions with dark-matter based models and implications on the nature of the Universe

Kroupa P., *Gjergo E.*, et al.; 2023

Proceedings of Corfu2022: Workshop on Tensions in Cosmology, 2023.

Software I developed

4. `pyIGIMF`: for the efficient computation of the initial mass function in the context of the IGIMF theory (public code)

Please refer to the Journal Article No. 19 below

3. `GalCEM`: A detailed Galactic Chemical Evolution (public code)

Please refer to the Journal Article No. 7 below

2. `Dust in Gadget3`: Dust Evolution within a complex zoom-in simulation (private proprietary code)

Please refer to the Journal Article No. 3 below

1. `Add-on`: Improved Photometric Type-Ia Supernova Estimator for the Dark Energy Survey and the Legacy Survey of Space and Time (LSST) (private proprietary code)

Please refer to the Journal Article No. 1 below

Journal Articles

26. Gravitational boost from second-order perturbations in a linearly expanding closed universe Gjergo, Kroupa (2026)

Physical Review D, published 13 November 2026

25. The MAGPI survey: emission-line products data release and the role of spectroscopic aperture covering fraction on the Balmer decrement–stellar mass relation

Battisti et al. (2026)

Monthly Notices of the Royal Astronomical Society, Volume 552, Issue 1, stag1585

24. The Sc, Ti, and V Abundance Discrepancy: Testing High-mass Initial Mass Function Variation and Massive-star Rotation

Choi, Gjergo, Kim, Kajino (2026)

The Astrophysical Journal, Volume 1008, Issue 1, id.126

Gjergo is the corresponding author

23. Limited imprint of high-mass initial mass function variations on sodium abundances in main sequence galaxies Guo et al. (2026)

Astronomy & Astrophysics, Volume 712, id.A96, 16 pp.

22. Baryonic mass budgets in the central regions of the Bullet Cluster and their consistency with strong lensing in MOND Zhang et al. (2026)

Physical Review D, Volume 114, Issue 2, id.023001, 21 pp.

21. The $m_{\rm max}$–$M_{\rm ecl}$ relation in the LEGUS clusters Zinnkann et al. (2026)

Astronomy & Astrophysics, Volume 709, id.A4, 17 pp.

20. The Initial Mass Function as the Equilibrium State of a Variational Process: Why the IMF Cannot be Sampled Stochastically

Gjergo, Zhang, Kroupa (2026)

Research in Astronomy and Astrophysics, Volume 26, Issue 5, id.055012, 12 pp.

19. Massive Star Formation at Supersolar Metallicities: Constraints on the Initial Mass Function Gjergo et al. (2026)

Research in Astronomy and Astrophysics, Volume 26, Issue 2, id.025003, 22 pp.

18. The effect of the environment-dependent stellar initial mass function on the baryonic Tully–Fisher relation Zonoozi et al. (2025)

Monthly Notices of the Royal Astronomical Society, Volume 542, Issue 3, pp. 1864-1872, 9 pp.

17. The impact of early massive galaxy formation on the cosmic microwave background Gjergo & Kroupa (2025)

Nuclear Physics, Section B, Volume 1017, id.116931

16. Alien Type Ia Supernovae from the Milky Way Merger History and One Possible Candidate: Kepler's Supernova He, Zhou, Gjergo, Fu (2025)

The Astrophysical Journal, Volume 986, Issue 2, id.123, 12 pp.

Gjergo is a corresponding author

15. The p-Laplacian as a framework for generalizing Newtonian gravity and Milgromian gravitation Scherer, Pflamm-Altenburg, Kroupa, Gjergo
Astronomy & Astrophysics, Volume 698, id.A167, 12 pp. (2025)

14. The MAGPI Survey – The kinematic morphology–density relation (or lack thereof) and the Hubble sequence at z ≈ 0.3 Foster et al. (2025)
Publications of the Astronomical Society of Australia, Volume 42, id.e058, 22 pp.

13. The effect of the environment-dependent stellar initial mass function on the photometric properties of star-forming galaxies
Astronomy & Astrophysics, Volume 689, id.A221, 18 pp. Haslbauer et al. (2024)

12. First Detection of CO Isotopologues in a High-redshift Main-sequence Galaxy: Evidence of a Top-heavy Stellar Initial Mass Function
The Astrophysical Journal, Volume 970, Issue 2, id.136, 11 pp. Guo et al. (2024)

11. The Variation in the Galaxy-wide Initial Mass Function for Low-mass Stars: Modeling and Observational Insights Yan et al. (2024)
The Astrophysical Journal, Volume 969, Issue 2, id.95, 11 pp.

10. α-enhanced Astrochemistry: The Carbon cycle in extreme galactic conditions Bisbas, Zhang, Gjergo, et al. (2024)
Monthly Notices of the Royal Astronomical Society, Volume 527, Issue 3, pp.8886-8906

9. Correction to: An analytical description of the evolution of binary orbital-parameter distributions in N-body computations of star clusters Kroupa, Wang, Gjergo (2023)
Monthly Notices of the Royal Astronomical Society, Volume 526, Issue 4, pp.5777-5777

8. Assessing stellar yields in Galaxy chemical evolution: Observational stellar abundance patterns Liang, Gjergo, Fan (2023)
Monthly Notices of the Royal Astronomical Society, Volume 522, Issue 1, pp.863-884
Gjergo is the corresponding author

7. GalCEM I: An Open-source Detailed Isotopic Chemical Evolution Code Gjergo et al. (2023)
The Astrophysical Journal Supplement Series, Volume 264, Issue 2, id.44, 22 pp.

6. LAMOST meets Gaia: The Galactic open clusters Fu et al. (2022)
Astronomy & Astrophysics, Volume 668, id.A4, 16 pp.

5. The MAGPI Survey – science goals, design, observing strategy, early results and theoretical framework Foster et al. (2021)
Publications of the Astronomical Society of Australia, 2021, Volume 38, article id. e031

4. On the origin of dust in galaxy clusters at low-to-intermediate redshift Gjergo et al. (2020)
Monthly Notices of the Royal Astronomical Society, Volume 493, Issue 2, p.2782-2792.

3. Dust evolution in galaxy cluster simulations Gjergo et al. (2018)
Monthly Notices of the Royal Astronomical Society, Volume 479, Issue 2, p.2588-2606.

2. Analytic photometric redshift estimator for Type Ia supernovae from LSST Wang, Gjergo, Kuhlmann (2015)
Monthly Notices of the Royal Astronomical Society, Volume 451, Issue 2, p.1955-1963.

1. Type Ia supernovae selection and forecast of cosmology constraints for the Dark Energy Survey Gjergo et al. (2013)
Astroparticle Physics, Volume 42, p. 52-61.

Invited Talks and Seminars

Origin of Elements and Cosmic Evolution (OECE2026, Beijing, CN)	Nov '26
Seminar at Cambridge University (Cambridge, UK)	Feb '26
Seminar at Bonn University (Bonn, Germany)	Feb '26
Seminar at Università la Sapienza (Rome, Italy)	Feb '26
Research group seminar at University of Florence (Florence, Italy)	Feb '26
Origin of Elements and Cosmic Evolution (OECE2025, Beijing, CN)	Nov '25
Seminar at Yunnan University (Kunming, China)	Jan '25
Seminar at IBS (Daejeon, South Korea)	Dec '24
Miniworkshop on GalCEM and GCE (Beijing, China)	July '24
Seminar at Beihang University (Beijing, China)	May '24
Origin of Elements and Cosmic Evolution (OECE2023, Beijing, CN)	Nov '23
Seminar at Tsinghua University (Beijing, China)	May '23
Seminar at Bonn University (Bonn, Germany)	Apr '22
Seminar at Charles University (Prague, Czech Republic)	Mar '22
GW NEXT 2022, Online Conference (Beijing, CN)	Jan '22
National Astronomical Observatory of China Seminar (Beijing, CN)	May '21
Kavli Institute for Astronomy & Astrophysics Seminar (Beijing, CN)	Nov '20
Beijing Normal University Seminar (Beijing, China)	Nov '19
3rd International Workshop on Physics (Tirana, Albania)	Oct '18
2nd International Workshop on Physics (Tirana, Albania)	Sep '16
CARA outreach (Yerkes Observatory, WI, USA)	Jul '13

Technical Talks

MAGPI Busyweek (online, Australia)	June '21
Hydrodynamical Simulations Meeting (Trieste, Italy)	Sep '16
LSST-DESC Collaboration Meeting (Pittsburgh, PA)	Dec '13

LSST DESC Supernovae Working Group (11 Webinars) Spring ‘17-‘18
DESSN Workshop (KICP at the University of Chicago, IL) Jul ‘13

Talks and posters

17th Zhang Heng CAS Academic Symposium (Xiamen, China) Jul ‘26

Galaxies at Crossroads (Brno, Czech Republic) Sep ‘24
CosmoChem2024 (Palinuro, Italy) Jun ‘24
Nuclei in the Cosmos (NIC XVII, Daejeon, South Korea) Sep ‘23
Origin of Matter and Evolution of Galaxies OMEG16 (Hanoi, Vietnam) Oct ‘22
Interstellar Physics & Chemistry (Online conference, Zhuhai, China) Jan ‘22
Origin of Elements and Cosmic Evolution (Beijing, China) Nov ‘19
The Milky Way: LAMOST and other leading surveys (Yichang, China) Oct ‘19
Second Italian Cluster Conference (Naples, Italy) May ‘18
First Italian Cluster Conference (Turin, Italy) Feb ‘17
KROME Computational school (Florence, Italy) Jun ‘16
ICTP Workshop on Large-Scale Structure (Trieste, Italy) Jun ‘16
ICTP Summer School of Cosmology (Trieste, Italy) Jun ‘16
Lucchin astrophysics school (Naples, Italy) May ‘16
AAS 223rd Meeting (Washington, D.C.) **Chambliss Hon. Mention** Jan ‘14
Santa Fe Cosmology Workshop (Santa Fe, NM) Jul ‘13
AAS 222nd Meeting (Indianapolis, IN) **Chambliss winner** Jun ‘13
AAS 221st Meeting (Long Beach, CA) Jan ‘13
ICTP Summer School of Cosmology (Trieste, Italy) Jun ‘12
Chicagoland and Midwest 1-Day Dark Matter Workshop (Chicago, IL) Apr ‘12
APS April Meeting (Atlanta, GA) Apr ‘12